ORIGINAL RESEARCH REPORT

# Integrated Bulk RNA-seq Meta-analysis Reveals Recurrent Transcriptional Signatures Across Human Cardiomyopathy Cohorts

*A journal-style technical report based on GSE71613 and GSE55296*

**Prepared by**

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Datasets: GSE71613 and GSE55296
Analysis framework: GREIN-derived counts, DESeq2, SVA, MetaVolcanoR, clusterProfiler, ReactomePA and STRING
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**Report status**

This document is a professional journal-style technical report of the current GitHub workflow. It has not undergone peer review. Gene-level and pathway-level conclusions are provisional until identifier harmonization and study-level count discrepancies are resolved.

**Code repository**

github.com/drsumayamifty/Cardiomyopathy_bulk_RNAseq_meta_analysis

# Abstract


**Structured summary**

Background: Cardiomyopathy and end-stage heart failure arise from heterogeneous etiologies, yet recurrent transcriptional changes may be detectable across independent myocardial RNA-seq cohorts. Methods: Public human cardiac RNA-seq datasets GSE71613 and GSE55296 were reanalyzed as a broad cardiomyopathy-versus-healthy comparison. The combined cohort comprised 44 samples: 30 cardiomyopathy samples and 14 healthy controls. GREIN-derived count matrices were filtered and analyzed separately with DESeq2. Surrogate variable analysis was incorporated into study-level designs, and a random-effects model was applied to gene-level log2 fold changes using MetaVolcanoR. Functional interpretation used Gene Ontology biological process, KEGG and Hallmark GSEA; over-representation analysis included GO, KEGG and Reactome. A STRING-based protein association network was summarized by degree centrality. Results: The pooled random-effects output contained 4,052 genes. Using an adjusted pooled P value below 0.05 and an absolute pooled log2 fold change of at least 1, 322 genes were retained (209 downregulated and 113 upregulated). The nominal-P volcano classification contained 225 downregulated and 128 upregulated genes. The project Venn diagram displayed 19 genes shared between the study-level DEG sets. GSEA showed nominal signals involving carbohydrate regulation, chromosome segregation, TGF-beta signaling, apical junction, mTORC1 signaling and selected metabolic pathways, but no reported pathway remained significant after multiple-testing correction. ORA produced no significant terms under the adjusted input-gene threshold. APP, CREBBP, BRCA1, CD4 and H4C6 ranked among the most connected network nodes. Conclusions: The workflow detects a broad, cross-cohort transcriptional pattern in cardiomyopathy, but the biological findings should be treated as exploratory. Identifier processing and inconsistent study-level DEG counts require correction before manuscript submission or mechanistic interpretation.


**Keywords:** cardiomyopathy; heart failure; bulk RNA sequencing; differential expression; meta-analysis; GSEA; protein interaction network

## Research highlights

- Two independent human cardiac RNA-seq cohorts were integrated across 44 samples.
- The random-effects analysis retained 322 FDR-adjusted genes at |pooled log2FC| >= 1.
- Pathway results were nominal rather than FDR-significant, and ORA returned no significant terms.
- Network centrality identified APP, CREBBP and BRCA1 as highly connected candidates.
- A critical gene-identifier harmonization issue must be resolved before publication.

**Critical analytical validation note**

The current annotation script removes all characters after the first period from Gene_ID values under the assumption that they are versioned Ensembl identifiers. However, the observed identifiers are numeric strings such as 20.5441 rather than canonical ENSG identifiers. This transformation can change the identity of a feature. Because the transformed IDs feed the annotation, Venn, meta-analysis, enrichment and network stages, the downstream gene-level results in this report are provisional. The original GREIN identifier definition should be confirmed, then all downstream analyses should be rerun from the unmodified character IDs.

# 1. Introduction

Cardiomyopathy encompasses structurally and functionally distinct myocardial disorders that can converge on ventricular dysfunction and end-stage heart failure. Although dilated, ischemic and restrictive phenotypes have different initiating mechanisms, shared transcriptional responses may reflect common processes such as altered metabolism, stress signaling, extracellular remodeling, cell-cycle regulation and changes in tissue composition.

Public RNA-seq repositories make it possible to test whether signals observed in one cohort recur in another. However, direct pooling of raw counts across experiments is often inappropriate because sequencing platforms, cohort composition and technical characteristics differ. A study-wise analysis followed by effect-size meta-analysis preserves each experiment's internal comparison while seeking a cross-study consensus.

This report documents a reproducible bulk RNA-seq workflow applied to GSE71613 and GSE55296. GSE71613 contains healthy controls and restrictive or dilated cardiomyopathy samples, whereas GSE55296 contains healthy controls and ischemic or dilated cardiomyopathy samples. The analysis therefore addresses a broad cardiomyopathy/heart-failure contrast rather than a subtype-specific dilated cardiomyopathy question. The objectives were to: (i) assess sample structure by PCA; (ii) identify study-level DEGs and overlap; (iii) estimate pooled gene effects with a random-effects model; (iv) explore functional pathways; and (v) identify highly connected proteins in a candidate interaction network.

# 2. Materials and Methods

## 2.1 Data sources and study design

The analysis used two public human cardiac RNA-seq datasets from the NCBI Gene Expression Omnibus. Dataset characteristics were verified against the GEO records and original publications [1-4]. All disease samples were collapsed into a single Cardiomyopathy group, and healthy donor samples formed the reference group.

**Table 1. Study characteristics**

| Dataset | Total n | Healthy | Disease | Disease composition | Platform | Analysis note |
|---|---|---|---|---|---|---|
| GSE71613 | 8 | 4 | 4 | 2 dilated; 2 restrictive | Illumina HiSeq 2000 | Human cardiac tissue |
| GSE55296 | 36 | 10 | 26 | 13 dilated; 13 ischemic | AB 5500xl Genetic Analyzer | Human cardiac tissue |
| Combined | 44 | 14 | 30 | Mixed cardiomyopathy etiologies | Two platforms | Study-wise analysis; no raw-count pooling |

## 2.2 Count matrices, metadata and preprocessing

Count matrices were obtained from the GREIN processing ecosystem, which provides uniformly processed GEO RNA-seq resources [5]. Each dataset was imported separately. Duplicate feature identifiers were removed before conversion to an integer matrix. Metadata rows were matched to sample columns, and sample labels containing normal, control or healthy terms were assigned to the healthy reference level. All other samples were assigned to Cardiomyopathy.

Low-expression features were retained when at least 10 reads were observed in at least half of the samples in the dataset. This filter reduced the influence of extremely sparse features while preserving genes measured across a meaningful fraction of the cohort.

## 2.3 Surrogate-variable modeling and differential expression

For each dataset, a full model containing the condition term and a null model containing only an intercept were used to estimate surrogate variables with the sva package [6]. Estimated surrogate variables were added to the DESeq2 design together with condition. DESeq2 modeled counts with a negative-binomial framework and provided log2 fold changes, standard errors, Wald statistics, raw P values and Benjamini-Hochberg adjusted P values [7,8]. With healthy as the reference, a positive log2 fold change indicates higher expression in the pooled cardiomyopathy group.

Study-level DEGs used for the overlap analysis were defined by adjusted $P < 0.05$ and $|\log2 \text{ fold change}| > 1$. Because GSE71613 contains only eight samples, surrogate-variable estimates and study-specific effect sizes should be considered less stable than those from GSE55296.

## 2.4 Principal component analysis

PCA was generated from variance-stabilized expression. The project labels the plots as pre-correction and post-correction. Technically, the first plot uses a blind variance-stabilizing transformation, whereas the second uses a design-aware transformation from the surrogate-variable-informed DESeq2 object. The latter is useful for visualizing design-associated structure but is not equivalent to explicitly removing batch effects from the expression matrix.

## 2.5 Annotation and overlap analysis

The annotation workflow used genekitr to associate feature identifiers with gene symbols, descriptions and biotypes, then retained protein-coding genes. Venn analysis used the annotated study-level tables and identified genes satisfying the DEG thresholds in each dataset. The intersection was exported as the set shared by all included studies.

## 2.6 Random-effects meta-analysis and volcano visualization

Study-wise differential-expression results were supplied to MetaVolcanoR [9]. The DESeq2 log2 fold-change standard error was squared to obtain within-study variance. A random-effects model produced a pooled effect (randomSummary), confidence interval and pooled P value for each jointly analyzed feature. Pooled P values were adjusted with the Benjamini-Hochberg method. The principal FDR-filtered result used randomP.adjust < 0.05 and |randomSummary| >= 1.

The custom volcano plot used the pooled effect on the x-axis and -log10(raw pooled P value) on the y-axis. Its color classification therefore reflects nominal P < 0.05 and |pooled log2FC| > 1 rather than the FDR-filtered gene set. The report keeps these two summaries separate.

## 2.7 Functional enrichment

Gene set enrichment analysis used the complete ranked list, with ranking score defined as sign(pooled log2FC) × -log10(pooled P value). GO Biological Process and KEGG GSEA were run with clusterProfiler, and Hallmark GSEA used MSigDB gene sets through msigdbr [10,11]. Positive normalized enrichment scores indicate concentration toward the cardiomyopathy-upregulated end of the ranked list; negative scores indicate concentration toward the downregulated end.

Over-representation analysis tested significant genes against GO, KEGG and Reactome using clusterProfiler and ReactomePA [11,12]. Empty result objects were not plotted. In the current run, no ORA terms passed the applied thresholds.

## 2.8 Protein-association network analysis

Candidate genes were mapped to a STRING protein-association network [13]. Degree, betweenness, closeness and eigenvector centrality were computed. The displayed hub-gene plot ranks nodes by degree centrality and colors bars by pooled log2 fold change. Network centrality indicates connectivity within the database-derived graph; it does not establish causality, tissue specificity or therapeutic actionability.

## 2.9 Reproducibility

The workflow is organized as sequential R scripts covering setup, DESeq2 analysis, annotation, Venn overlap, meta-analysis, custom volcano plotting, enrichment and network analysis. Code and generated outputs are hosted in the public GitHub repository listed on the title page. A final publication-ready release should also record package versions with sessionInfo() or an renv lockfile.

# 3. Results

## 3.1 Cohort structure and PCA

The combined analysis represented 44 human cardiac samples: 14 healthy controls and 30 cardiomyopathy samples. The larger GSE55296 cohort dominated the total sample count and contained both dilated and ischemic cardiomyopathy, whereas GSE71613 contributed only four disease samples and four controls.

In GSE55296, the blind VST PCA showed substantial overlap and dispersion across the first two components. In the design-aware PCA, PC1 explained a larger fraction of variance and healthy samples tended to occupy higher PC2 values, although overlap remained. GSE71613 showed partial condition separation primarily along PC1, but the very small cohort and within-group dispersion limit confidence.

## Figure 1. Principal component structure

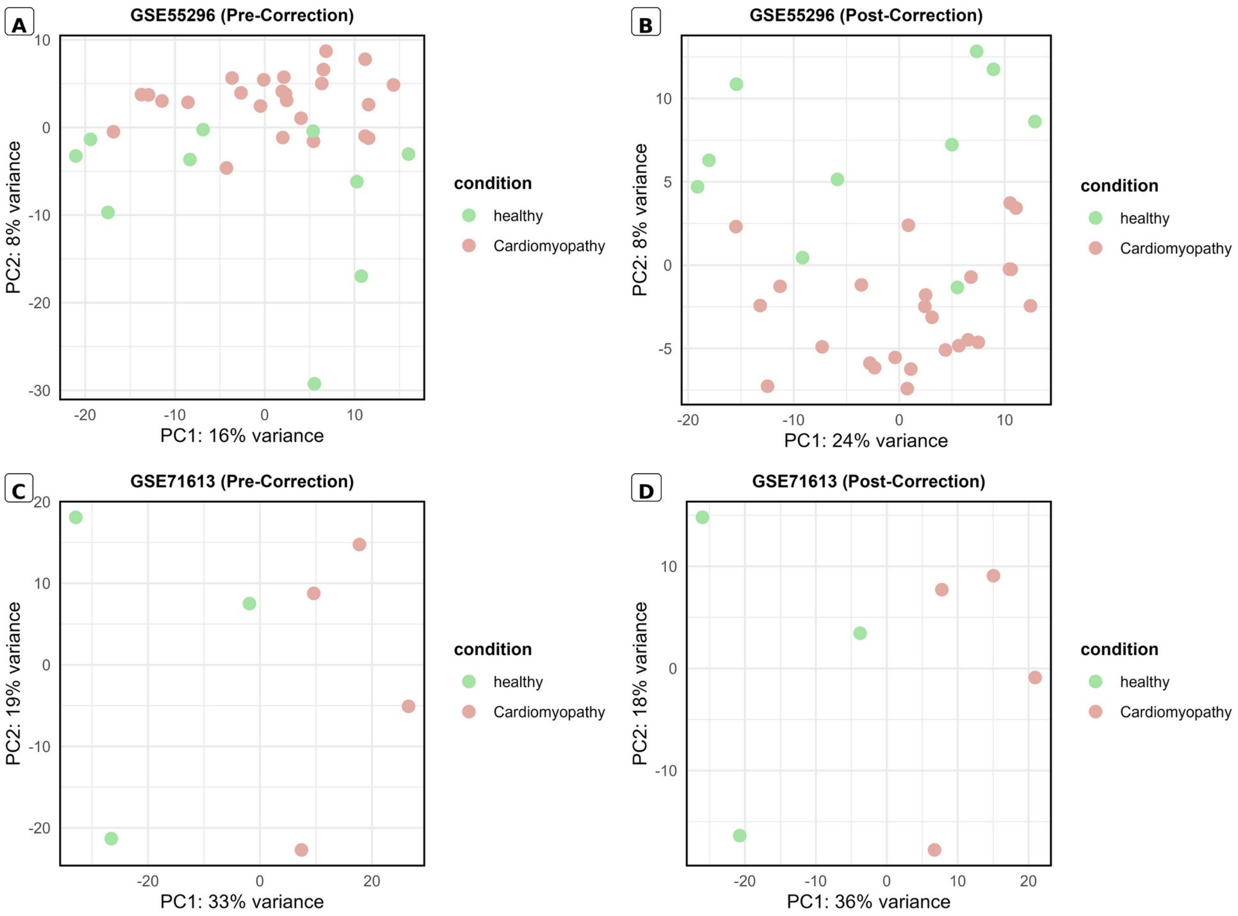


**Figure 1.** Principal component analysis of variance-stabilized expression. (A) GSE55296 blind VST: PC1 16%, PC2 8%. (B) GSE55296 design-aware VST: PC1 24%, PC2 8%. (C) GSE71613 blind VST: PC1 33%, PC2 19%. (D) GSE71613 design-aware VST: PC1 36%, PC2 18%. Green points are healthy samples and salmon points are cardiomyopathy samples.

**Interpretation.** The design-aware views emphasize condition-associated structure, particularly in GSE55296, but they do not show complete class separation. GSE71613 contains only eight samples, so the apparent separation is sensitive to individual samples and should not be interpreted as evidence of a robust classifier. The post-labeled panels are design-aware transformations rather than formally batch-corrected expression matrices.

## 3.2 Study-level differential expression and overlap

Using the annotated-table DEG threshold, the project Venn output displayed 354 genes in the GSE55296 set and 25 genes in the GSE71613 set. Nineteen genes were shared, corresponding to 335 GSE55296-only genes and six GSE71613-only genes.

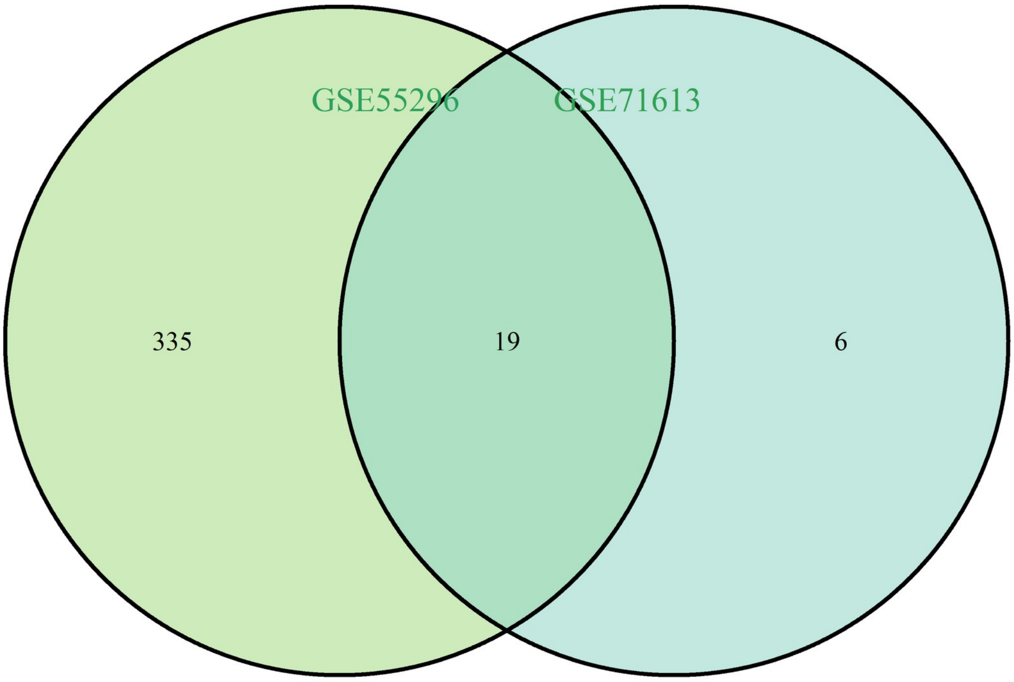


**Figure 2.** Overlap of study-level DEGs at adjusted P < 0.05 and |log2FC| > 1. The project figure reports 335 genes unique to GSE55296, six unique to GSE71613 and 19 shared genes.

**Interpretation.** The 19-gene intersection represents recurrence under the study-level threshold, not a formal meta-analysis. A gene can enter an overlap even when effect magnitudes differ, and direction agreement should be verified explicitly. Importantly, the raw DESeq2 result files contain 40 qualifying rows for GSE55296 and 677 for GSE71613, which does not match the counts displayed here. This discrepancy must be reconciled by tracing identifier conversion, annotation filtering and dataset provenance before the Venn result is used in a manuscript.

## 3.3 Random-effects differential-expression meta-analysis

The random-effects table contained 4,052 pooled features. The FDR-filtered output retained 322 genes at randomP.adjust < 0.05 and |pooled log2FC| >= 1. Of these, 209 had negative pooled effects and 113 had positive pooled effects. The imbalance toward negative effects suggests that the retained consensus signature was weighted toward lower expression in cardiomyopathy, subject to the identifier validation caveat.

**Table 2. Summary of pooled differential-expression results**

| Metric | Count | Definition |
|---|---|---|
| Genes entering pooled result | 4,052 | All features in random_effect_model.csv |
| FDR-filtered pooled DEGs | 322 | randomP.adjust < 0.05 and |randomSummary| >= 1 |
| FDR-filtered downregulated | 209 | Negative pooled log2FC |
| FDR-filtered upregulated | 113 | Positive pooled log2FC |
| Nominal volcano downregulated | 225 | raw pooled P < 0.05 and pooled log2FC < -1 |
| Nominal volcano upregulated | 128 | raw pooled P < 0.05 and pooled log2FC > 1 |

**Table 3. Representative annotated genes in the FDR-filtered pooled result**

| Gene | Direction | Pooled log2FC | 95% CI | FDR |
|---|---|---|---|---|
| CLCN4 | Down | -4.47 | -5.45 to -3.48 | 2.95e-16 |
| RUNX1 | Down | -3.83 | -4.55 to -3.11 | 1.65e-22 |
| ALDOA | Down | -4.32 | -5.38 to -3.25 | 4.20e-13 |
| CA8 | Down | -3.76 | -4.61 to -2.91 | 1.20e-15 |
| AMFR | Down | -3.65 | -4.47 to -2.83 | 8.18e-16 |
| TUBG1 | Down | -2.66 | -2.97 to -2.35 | 1.44e-59 |

| Gene | Direction | Pooled log2FC | 95% CI | FDR |
|---|---|---|---|---|
| BMPR2 | Down | -2.23 | -2.73 to -1.74 | 6.03e-16 |
| BRCA1 | Down | -2.41 | -3.08 to -1.73 | 6.11e-10 |
| BCKDHB | Up | 4.53 | 2.96 to 6.10 | 1.10e-06 |
| GALR2 | Up | 3.35 | 2.24 to 4.46 | 3.10e-07 |
| HERC3 | Up | 1.96 | 1.54 to 2.39 | 1.41e-16 |
| HPS1 | Up | 2.39 | 1.55 to 3.22 | 1.47e-06 |
| CA7 | Up | 2.03 | 1.38 to 2.69 | 1.18e-07 |

Representative candidates included lower pooled expression of CLCN4, RUNX1, ALDOA, CA8, AMFR, TUBG1, BMPR2 and BRCA1, together with higher pooled expression of BCKDHB, GALR2, HERC3, HPS1 and CA7. These genes span ion transport, transcriptional regulation, metabolism, signaling and cellular organization. Their biological interpretation is provisional because annotation depends on the unresolved identifier transformation.

## Figure 3. Pooled meta-analysis volcano plot

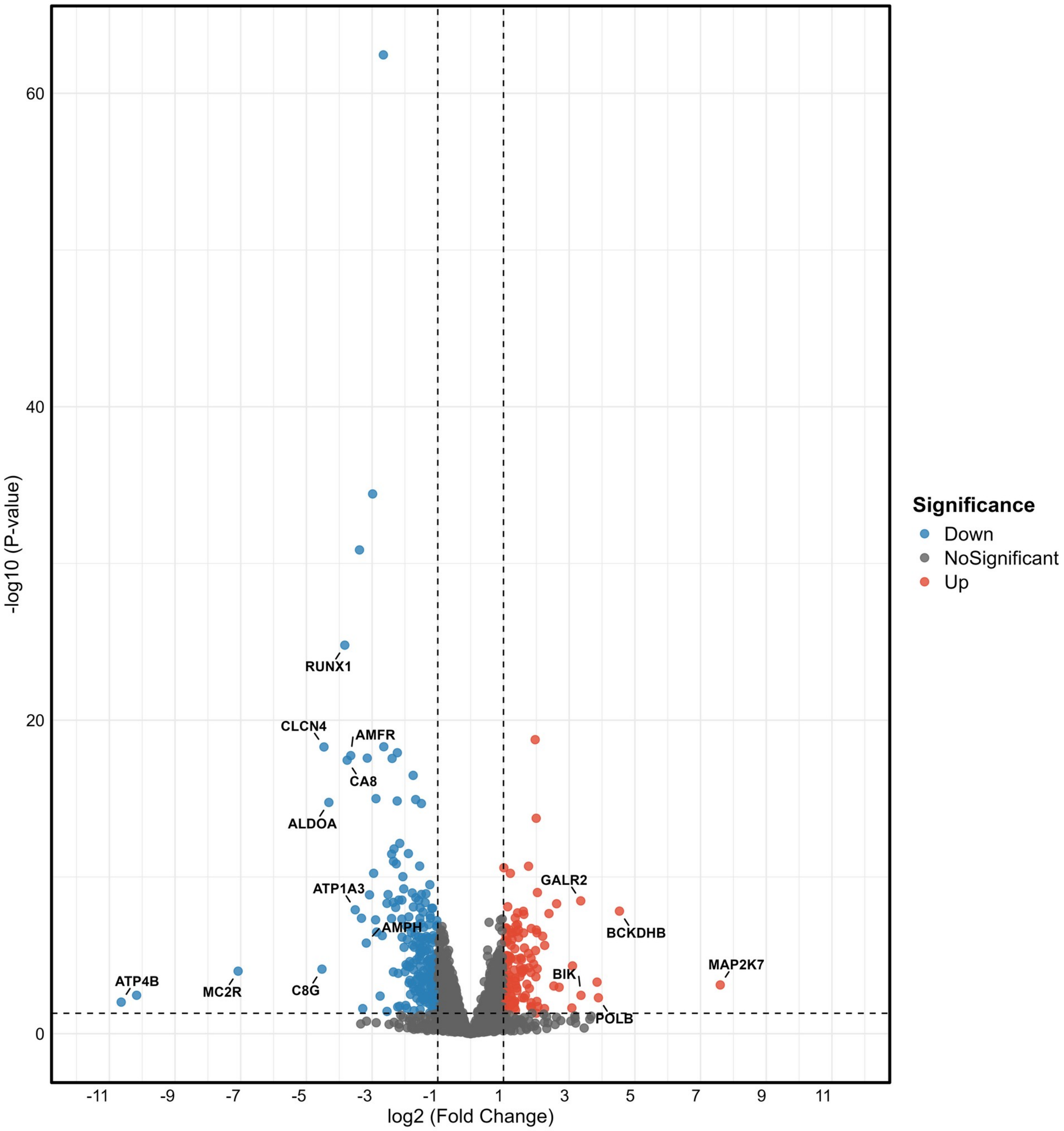


**Figure 3.** Volcano plot of random-effects pooled expression. The x-axis shows pooled log2 fold change and the y-axis shows -log10(raw pooled P value). Vertical dashed lines mark log2FC = -1 and +1; the horizontal dashed line marks P = 0.05. The nominal classification contains 225 downregulated genes (blue) and 128 upregulated genes (red); grey points do not meet both thresholds. Labels identify selected high-effect genes.

**Interpretation.** The plot reveals a broader nominal signal than the FDR-filtered result because its y-axis and color assignment use raw P values. Extreme pooled effects, including ATP4B and MAP2K7, require careful inspection of study-specific expression, standard errors and outliers. The volcano should not be described as an FDR-significant plot unless it is rebuilt with randomP.adjust.

### 3.4 Functional enrichment

The GSEA tables contained nominally enriched gene sets, but every displayed pathway had an adjusted P value above 0.05. Accordingly, the pathway patterns are hypothesis-generating rather than statistically confirmed. ORA returned zero significant pathways for the adjusted input-gene sets, so ORA plots were correctly not produced.

**Table 4. Selected exploratory GSEA signals**

| Collection | Gene set | NES | Nominal P | FDR |
|---|---|---|---|---|
| GO BP | Negative regulation of carbohydrate metabolic process | 2.02 | 0.00129 | 0.512 |
| GO BP | Mitotic sister chromatid segregation | -1.98 | 0.000139 | 0.440 |
| GO BP | TGF-beta receptor superfamily signaling | -1.88 | 0.00122 | 0.512 |
| Hallmark | Androgen response | 1.83 | 0.00493 | 0.123 |
| Hallmark | Apical junction | -1.75 | 0.00368 | 0.123 |
| Hallmark | mTORC1 signaling | -1.71 | 0.00916 | 0.153 |
| Hallmark | TGF-beta signaling | -1.59 | 0.0224 | 0.224 |
| KEGG | Fructose and mannose metabolism | -1.72 | 0.00205 | 0.720 |
| KEGG | Synaptic vesicle cycle | -1.68 | 0.0112 | 0.746 |
| KEGG | TGF-beta signaling pathway | -1.64 | 0.0128 | 0.746 |

Positive NES values indicate enrichment toward the cardiomyopathy-upregulated end of the ranked list, whereas negative values indicate enrichment toward the downregulated end. The lowest Hallmark FDR was approximately 0.123, and the GO and KEGG FDR values were higher; none met the conventional 0.05 threshold.

## Figure 4. GO Biological Process GSEA dot plot

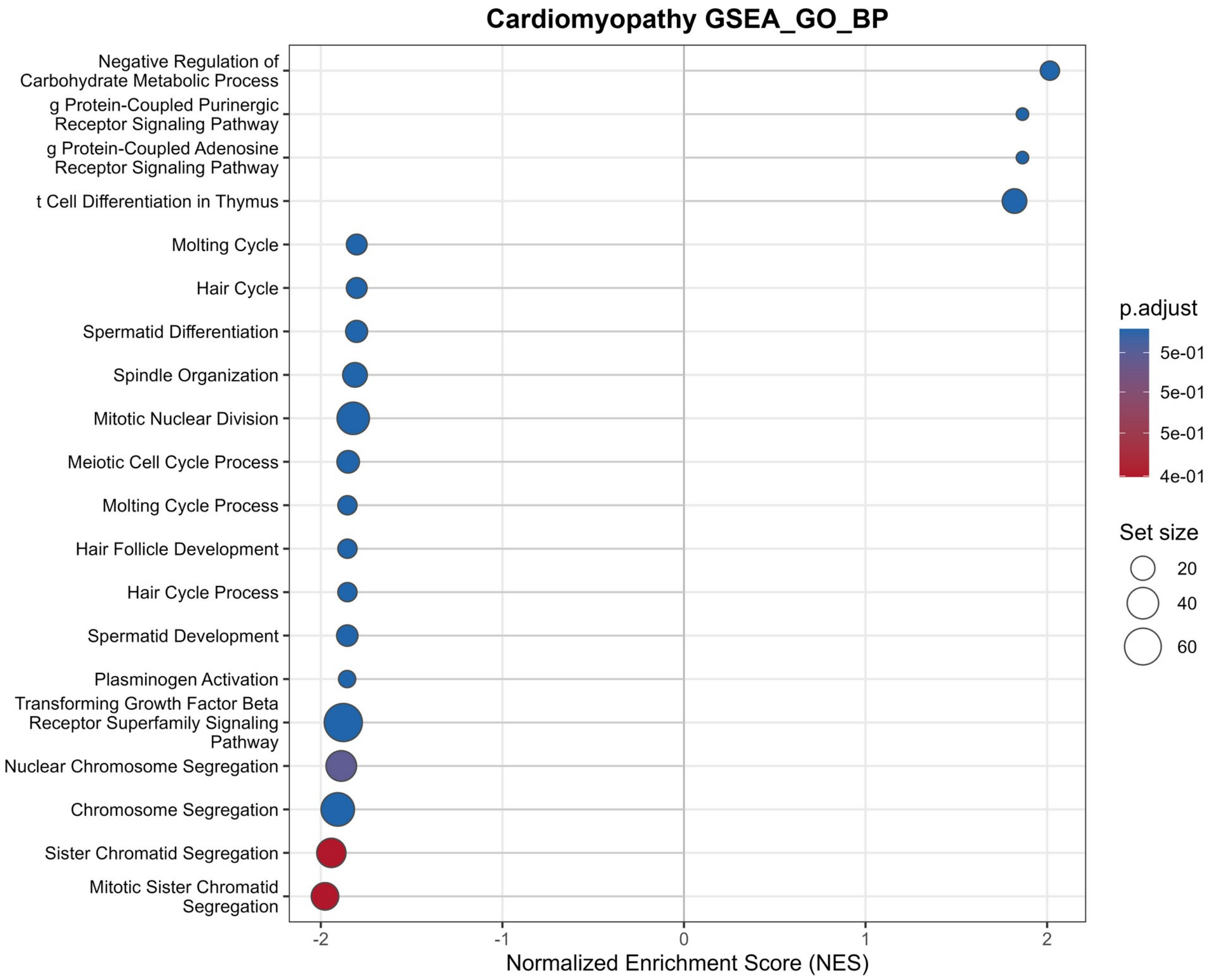


**Figure 4.** GO Biological Process GSEA summary. The horizontal axis is normalized enrichment score. Dot size represents gene-set size and dot color represents adjusted P value. Positive signals include negative regulation of carbohydrate metabolic process and purinergic/adenosine receptor signaling, whereas negative signals include chromosome segregation, mitotic sister chromatid segregation and TGF-beta receptor superfamily signaling.

**Interpretation.** The GO profile suggests possible shifts in metabolic regulation and reduced representation of chromosome-segregation and TGF-beta-related gene sets. However, the leading GO terms had FDR values around 0.44-0.51. They should therefore be reported as nominal patterns, not significant biological processes.

## Figure 5. GO Biological Process enrichment curves

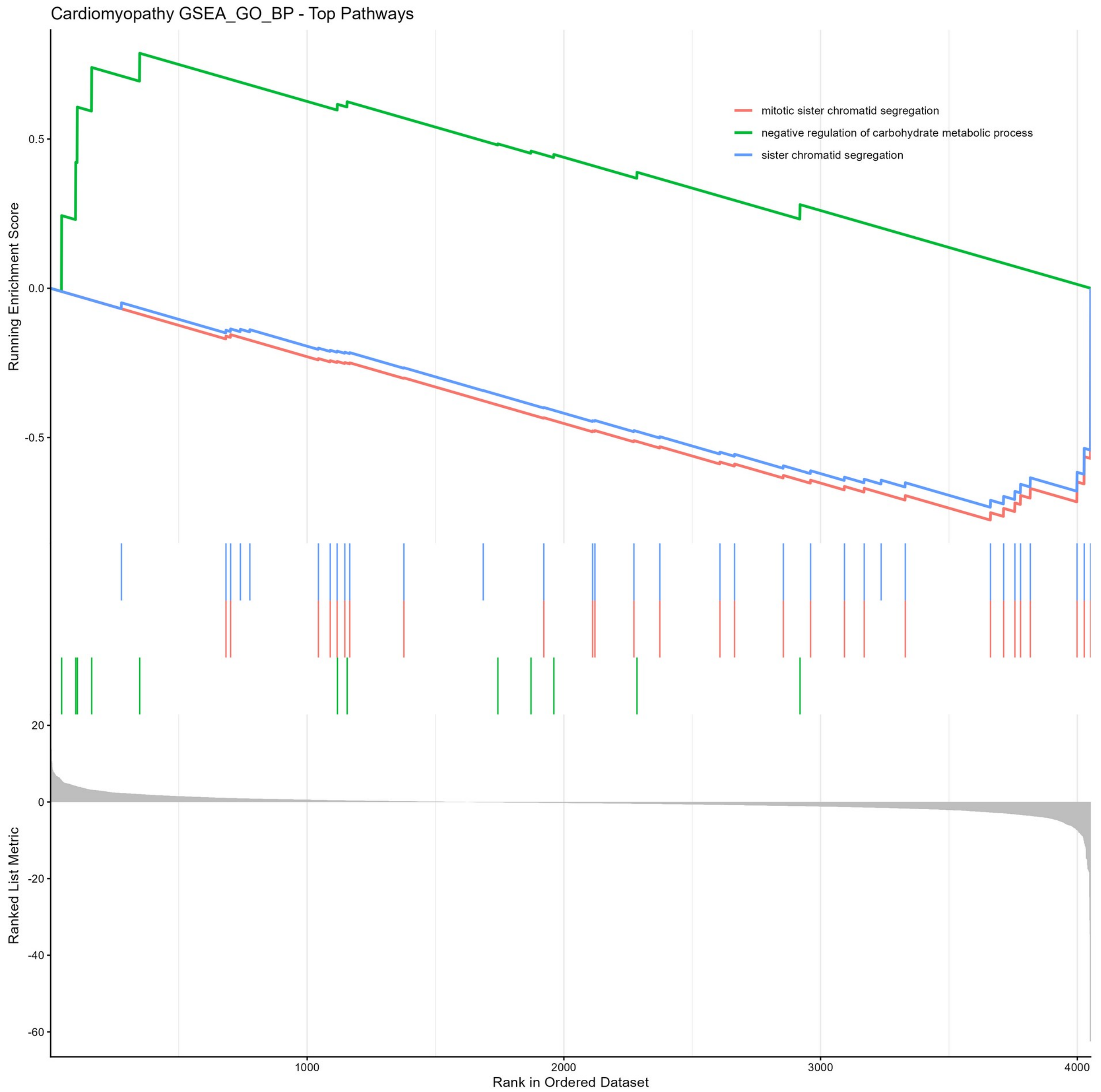


**Figure 5.** Running enrichment-score curves for the top GO Biological Process gene sets. Each colored line tracks the cumulative enrichment score across the ranked gene list. Vertical tick marks show positions of member genes, and the lower panel shows the ranked-list metric.

**Interpretation.** A curve rising above zero near the left side indicates that gene-set members cluster among genes with positive ranking scores; a curve falling below zero indicates clustering among negative scores. The curves visualize where enrichment arises but do not override the non-significant FDR values.

## Figure 6. Hallmark GSEA dot plot

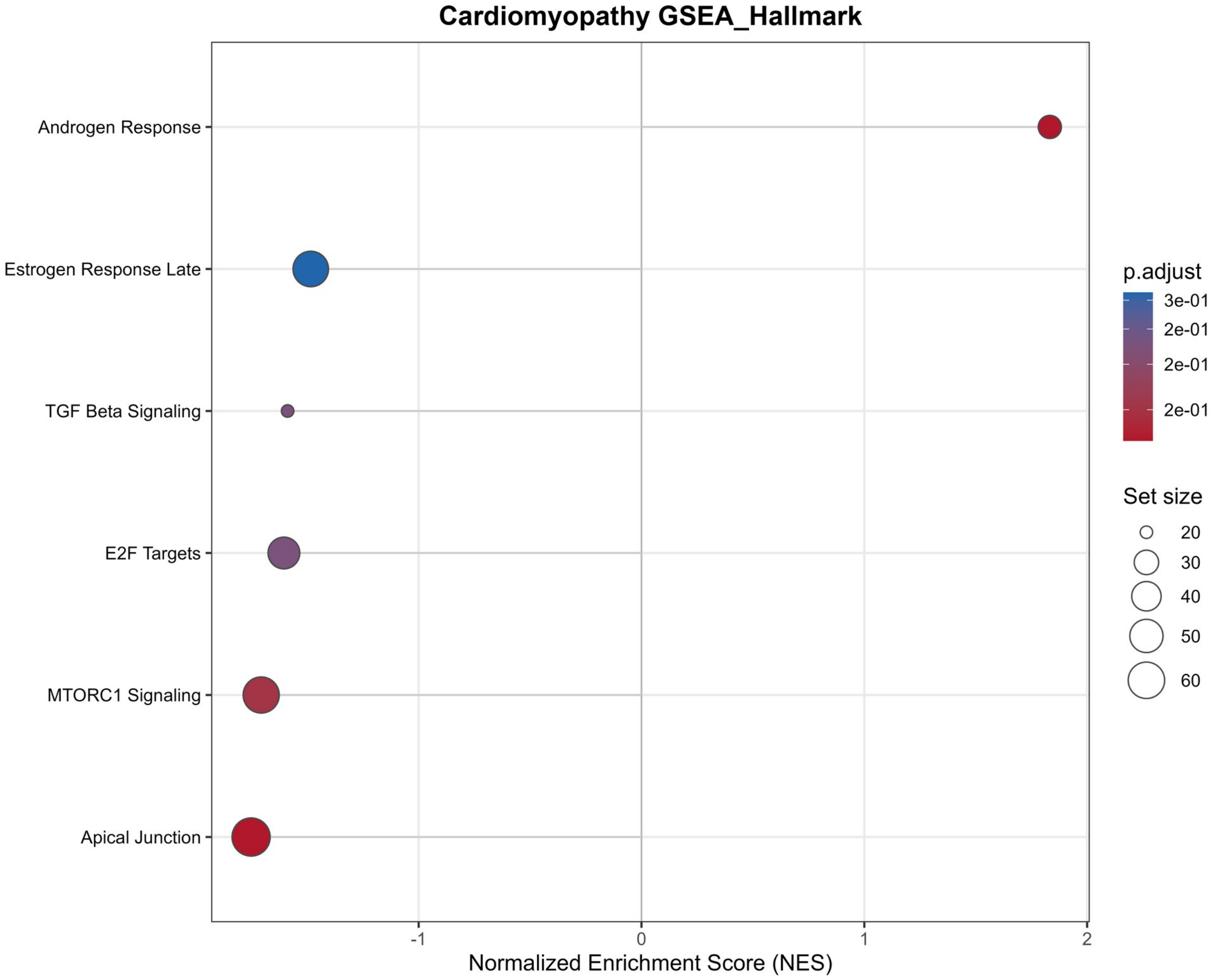


**Figure 6.** Hallmark GSEA summary. Androgen response showed positive enrichment (NES 1.83; nominal P 0.0049; FDR 0.123). Apical junction, mTORC1 signaling, E2F targets, TGF-beta signaling and late estrogen response showed negative enrichment.

**Interpretation.** The Hallmark pattern points toward coordinated changes in junctional organization, growth signaling, cell-cycle programs and hormonal-response gene sets. None passed FDR < 0.05, so the figure supports prioritization for follow-up rather than a claim of pathway activation or suppression.

## Figure 7. Hallmark enrichment curves

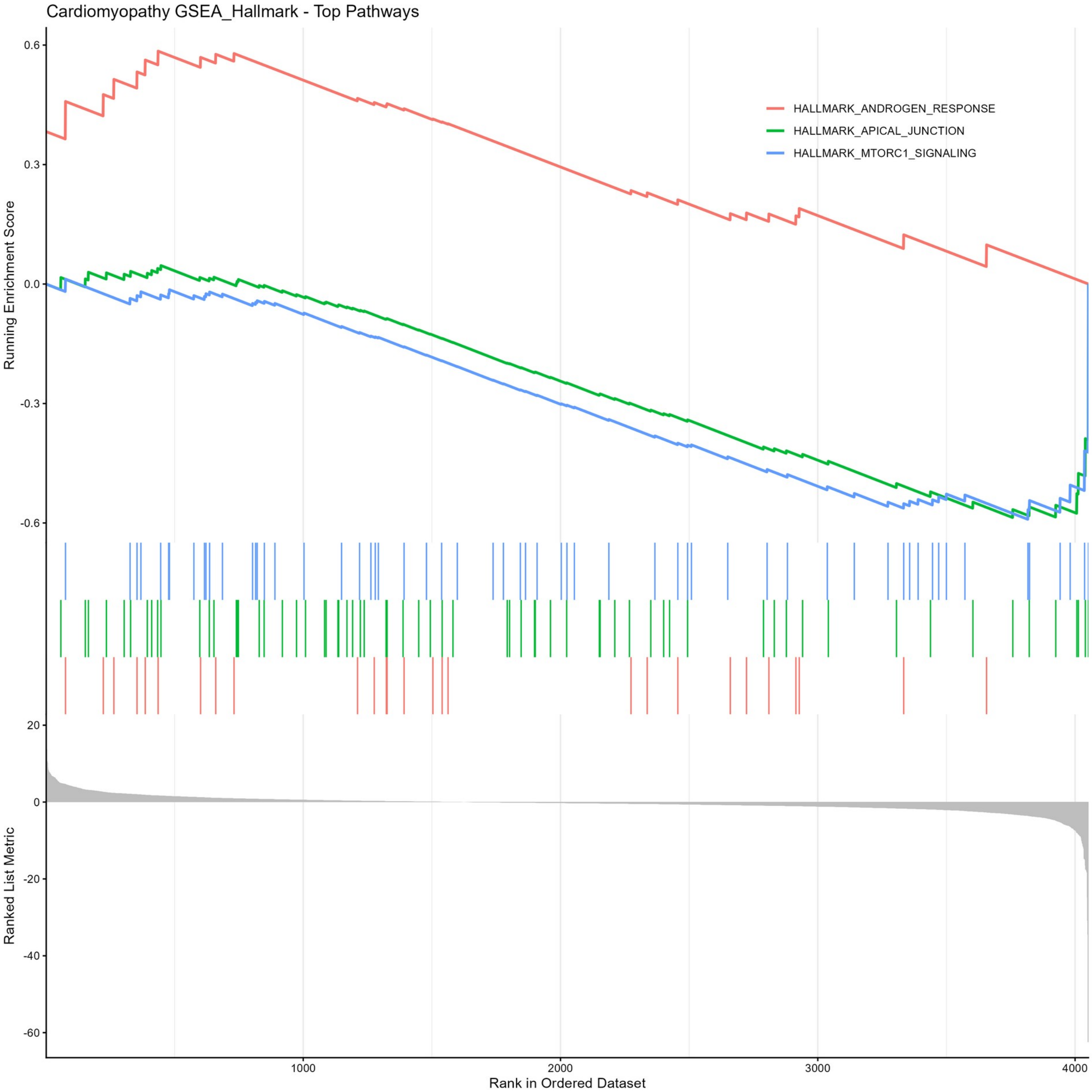


**Figure 7.** Running enrichment-score curves for the top Hallmark gene sets. The androgen-response curve is shifted toward the positive end of the ranking, whereas selected junctional and signaling programs are shifted toward the negative end.

**Interpretation.** The separation of positive and negative curves is consistent with the NES signs in the dot plot. Because multiple gene sets share member genes and all FDR values exceed 0.05, these curves should be interpreted collectively and cautiously.

## Figure 8. KEGG GSEA dot plot

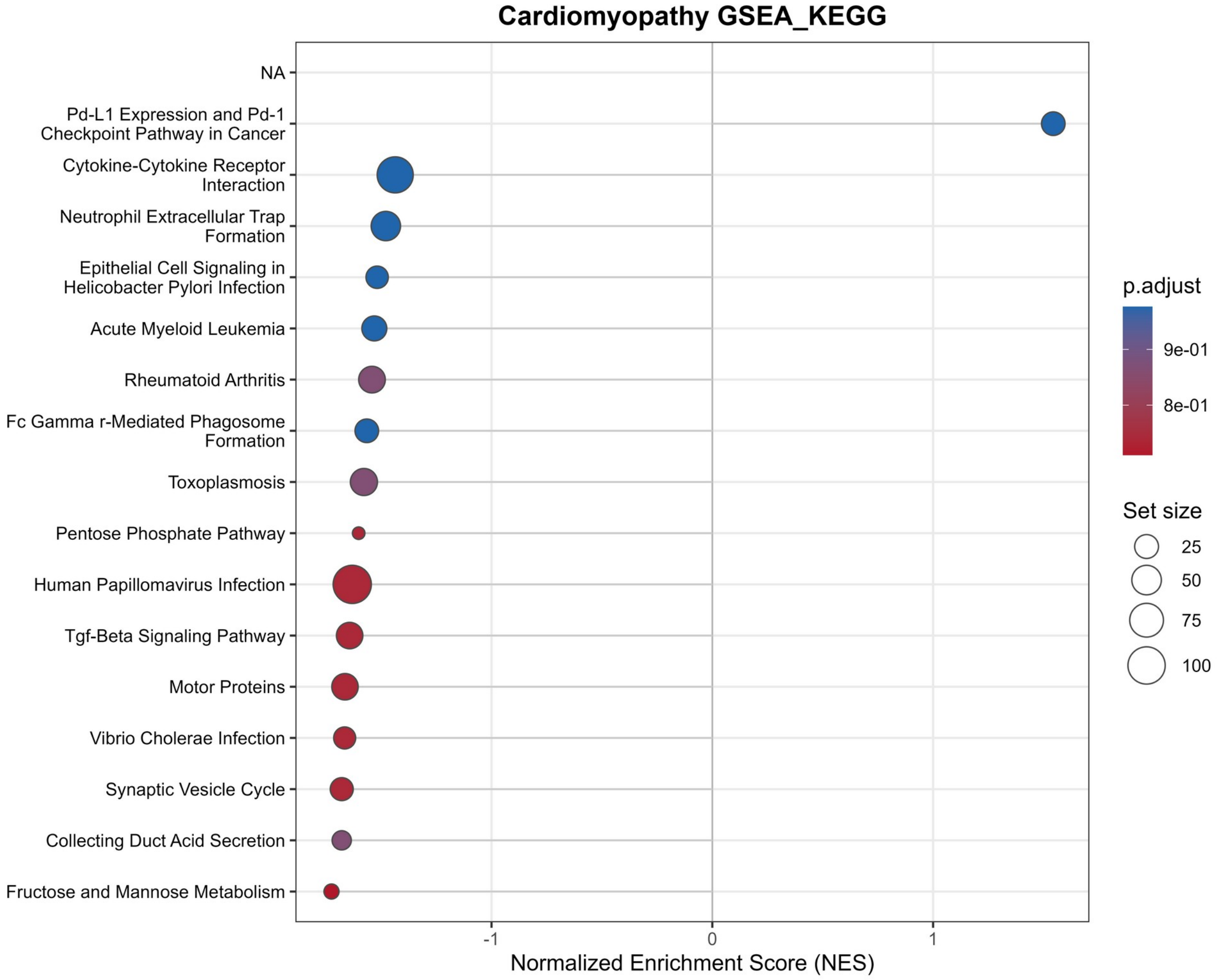


**Figure 8.** KEGG GSEA summary. Most displayed pathways had negative NES values, including fructose and mannose metabolism, collecting-duct acid secretion, synaptic vesicle cycle, motor proteins, pentose phosphate pathway and TGF-beta signaling. PD-L1 expression and PD-1 checkpoint pathway in cancer was the principal positive-NES pathway shown.

**Interpretation.** The negative pathway pattern may reflect broad changes in metabolic, vesicular, cytoskeletal and signaling genes in cardiomyopathy tissue. KEGG FDR values were high (the leading values were approximately 0.72 or greater), so the terms are exploratory and may also reflect annotation overlap rather than disease-specific pathway activity.

## Figure 9. KEGG enrichment curves

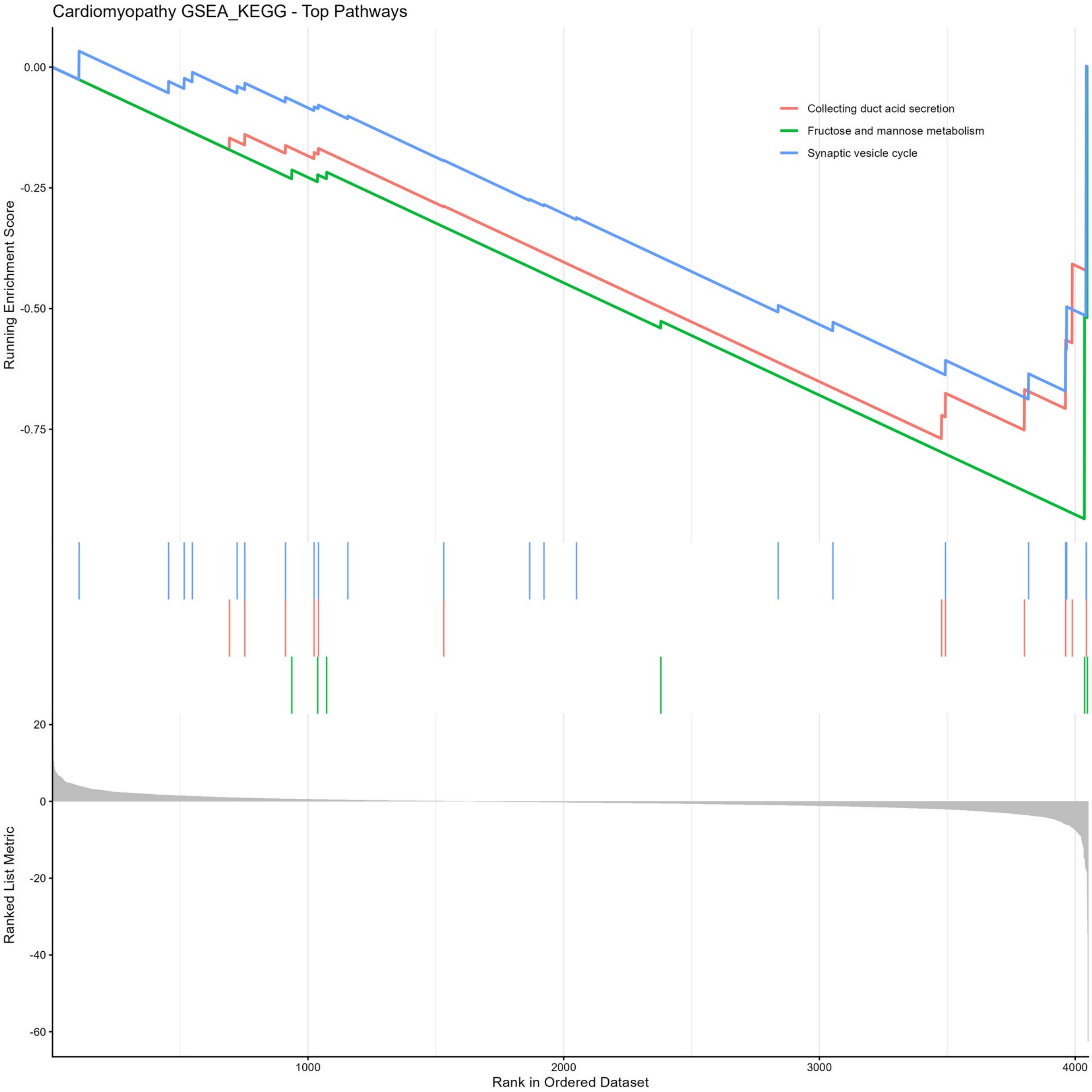


**Figure 9.** Running enrichment-score curves for the top KEGG pathways. Colored lines show cumulative enrichment across the ranked gene list, vertical ticks locate pathway genes, and the lower trace represents the ranking metric.

**Interpretation.** The curves demonstrate that the top KEGG signals are driven by genes concentrated toward the negative side of the ranking. Their utility is descriptive because no KEGG pathway achieved FDR significance.

## 3.5 Protein-association network and hub genes

The network analysis ranked 20 candidate genes by degree centrality. APP had the highest degree (13), followed by CREBBP and BRCA1 (11 each), CD4 (9), H4C6 (8) and E2F1 (7). The hub list combined both positively and negatively expressed genes.

**Table 5. Leading network hubs**

| Gene | Degree | Betweenness | Closeness | Pooled log2FC | Direction |
|---|---|---|---|---|---|
| APP | 13 | 0.254 | 0.236 | 1.11 | Up |
| CREBBP | 11 | 0.214 | 0.233 | -1.03 | Down |
| BRCA1 | 11 | 0.056 | 0.207 | -2.41 | Down |
| CD4 | 9 | 0.110 | 0.148 | 1.23 | Up |
| H4C6 | 8 | 0.042 | 0.199 | -1.08 | Down |
| E2F1 | 7 | 0.127 | 0.210 | -1.12 | Down |
| GSR | 6 | 0.130 | 0.141 | 1.15 | Up |
| AK1 | 6 | 0.070 | 0.115 | -1.36 | Down |
| ITGB1 | 6 | 0.231 | 0.225 | -1.26 | Down |
| BUB1B | 5 | 0.003 | 0.180 | -1.69 | Down |

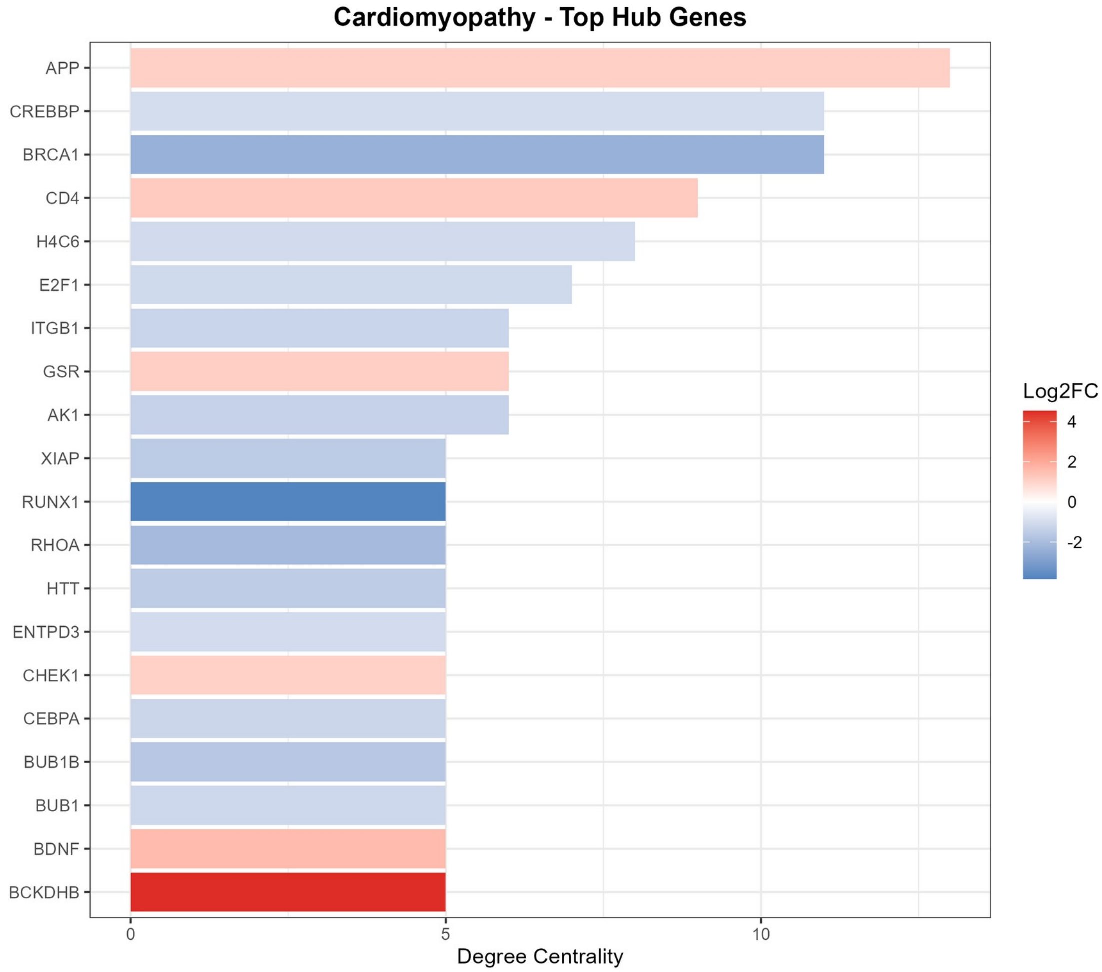


**Figure 10.** Top network hubs ranked by degree centrality. Bar length is the number of network connections. Bar color represents pooled log2 fold change, with warmer colors indicating positive effects and cooler colors indicating negative effects.

**Interpretation.** APP, CREBBP and BRCA1 occupy central network positions, but degree depends on the submitted gene set, STRING evidence coverage and interaction threshold. Highly connected proteins are not automatically the strongest biomarkers or best therapeutic targets. Independent expression validation and cell-type localization are required.

## 4. Discussion

This two-cohort analysis sought recurrent transcriptional changes across mixed cardiomyopathy phenotypes. The sample-level PCA results show that condition contributes to expression structure, especially in the larger GSE55296 cohort, but the classes are not fully separated. This is expected in human end-stage cardiac tissue, where etiology, treatment, age, ischemic injury, cell-type composition and technical factors can all influence bulk expression.

The pooled model produced 322 FDR-filtered genes with large effects, with more negative than positive pooled changes. Candidate genes span several broad functional categories. Lower pooled expression of RUNX1, BRCA1, TUBG1 and BMPR2 points to altered transcriptional, cell-cycle and signaling programs, while ALDOA, BCKDHB and CA-family genes highlight metabolic and ionic processes. These observations are descriptive because the current identifier pipeline has not been validated.

The pathway analysis is internally consistent with the absence of strong over-representation results. GSEA can detect distributed shifts across a ranked list even when no small DEG subset is strongly over-represented. Here, Hallmark, GO and KEGG analyses produced nominal signals, but adjustment for multiple testing removed statistical significance. The appropriate conclusion is that the data suggest candidate pathways for future testing, not that these pathways are confirmed as activated or suppressed.

The network analysis adds a systems-level view by highlighting nodes with many known or predicted protein associations. APP, CREBBP, BRCA1 and CD4 may be useful prioritization candidates, but network hubs often reflect extensive prior study and database coverage. Their centrality should be combined with effect consistency, cardiac cell-type expression, external cohorts and experimental validation.

A major strength is the transparent, script-based workflow and separation of study-level analyses before pooling. A second strength is the inclusion of effect sizes and uncertainty rather than relying solely on Venn overlap. The main weakness is data harmonization: a feature identifier error can propagate through every gene-level downstream step. Correcting that issue is more important than adding further visualization or pathway databases.

## 5. Limitations and Required Quality-Control Actions

**Table 6. Quality-control findings and corrective actions**

| No. | Issue | Required action / interpretation |
|---|---|---|
| 1 | Identifier definition | The current pipeline treats dotted numeric identifiers as versioned Ensembl IDs and truncates them at the first period. Confirm the original GREIN feature-ID schema and preserve IDs as character strings. |
| 2 | Inconsistent DEG counts | Raw DESeq2 tables yield 40 qualifying rows for GSE55296 and 677 for GSE71613, while the Venn figure displays 354 and 25. Regenerate annotation and overlap outputs after identifier correction. |
| 3 | Only two studies | Random-effects heterogeneity is poorly estimated with two cohorts. A heterogeneity P value of 1 should not be interpreted as proof that effects are homogeneous. |
| 4 | Clinical heterogeneity | Dilated, ischemic and restrictive cardiomyopathy were pooled. Results represent a broad cardiomyopathy/heart-failure signature and cannot be assigned to one subtype. |
| 5 | Small GSE71613 cohort | The dataset contains four controls and four cases. PCA, surrogate-variable estimation and gene-level variance estimates are sensitive to individual samples. |
| 6 | PCA terminology | The post-correction label describes a design-aware VST, not explicit removal of surrogate-variable effects from the assay. |
| 7 | Overlap direction | The 19-gene intersection should be checked for concordant log2 fold-change direction before it is called a shared disease signature. |
| 8 | Nominal pathway evidence | No GSEA collection achieved FDR < 0.05, and ORA returned zero significant terms. Pathways are exploratory. |

| No. | Issue | Required action / interpretation |
|---|---|---|
| 9 | Network inference | STRING interactions and degree centrality are database-derived and do not validate direct cardiac interactions or causal roles. |
| 10 | Software provenance | Record R and package versions, use renv, and rerun the entire workflow from a clean session before release. |

### Recommended validation sequence

1. Inspect the first column of the original GREIN count files and document the identifier type and delimiter.
2. Keep the original identifier as character. Split multi-gene identifiers only according to the source specification.
3. Map IDs with an explicit source key type, then report mapping rate and one-to-many mappings.
4. Regenerate DESeq2 tables, annotated tables, Venn overlap and the random-effects input.
5. Confirm that the same biological gene is matched across studies and check effect-direction concordance.
6. Rebuild GSEA, ORA and the STRING network from the corrected pooled result.
7. Validate leading genes in an independent human cardiac RNA-seq cohort or by targeted expression measurement.

## 6. Conclusions

The current workflow integrates two public human cardiac RNA-seq cohorts and provides a complete sequence of sample-level visualization, study-wise differential expression, random-effects pooling, pathway exploration and network prioritization. It reports 322 FDR-filtered pooled genes and recurrent nominal pathway patterns, while the network highlights APP, CREBBP and BRCA1 as highly connected candidates.

The scientifically defensible conclusion is that the project demonstrates a promising and reproducible analytical framework, not yet a validated cardiomyopathy gene signature. After correction of the identifier pipeline and reconciliation of the study-level DEG counts, the analysis can be rerun and upgraded into a manuscript-ready study with stronger gene-level and pathway-level claims.

## 7. Data and Code Availability

The source datasets are publicly available through NCBI GEO under accession numbers GSE71613 and GSE55296. The analysis scripts, tabular outputs and figures are available in the public repository: github.com/drsumayamifty/Cardiomyopathy_bulk_RNAseq_meta_analysis.

## 8. Ethics Statement

This secondary analysis used de-identified, publicly available transcriptomic datasets. No new human participants were recruited and no identifiable clinical data were accessed. Ethical approval and consent procedures for tissue collection were handled by the investigators of the original studies.

## 9. Author Contributions

Dr Sumaya Khan Mifty conceived the secondary analysis, developed and executed the R workflow, curated the outputs and prepared the GitHub repository. This report was generated from the documented project files and should be reviewed and edited by the author before journal submission.

## 10. Competing Interests

No competing interests were declared for this technical report.

## References

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# Appendix A. Figure File Inventory

| Report panel | Project filename | Content |
|---|---|---|
| Figure 1A | GSE55296_pre.png | Blind VST PCA |
| Figure 1B | GSE55296_post.png | Design-aware VST PCA |
| Figure 1C | GSE71613_pre.png | Blind VST PCA |
| Figure 1D | GSE71613_post.png | Design-aware VST PCA |
| Figure 2 | venn_diagram.png | Study-level DEG overlap |
| Figure 3 | volcano_plot.png | Random-effects volcano plot |
| Figure 4 | GSEA_GO_BP_dotplot.png | GO BP GSEA dot plot |
| Figure 5 | GSEA_GO_BP_enrichplot.png | GO BP enrichment curves |
| Figure 6 | GSEA_Hallmark_dotplot.png | Hallmark GSEA dot plot |
| Figure 7 | GSEA_Hallmark_enrichplot.png | Hallmark enrichment curves |
| Figure 8 | GSEA_KEGG_dotplot.png | KEGG GSEA dot plot |
| Figure 9 | GSEA_KEGG_enrichplot.png | KEGG enrichment curves |
| Figure 10 | hub_genes_barplot.png | Top network hubs |

# Appendix B. Interpretation Rules Used in This Report

| Term | Interpretation |
|---|---|
| Positive log2FC / NES | Higher expression or enrichment toward the cardiomyopathy-upregulated end. |
| Negative log2FC / NES | Lower expression or enrichment toward the cardiomyopathy-downregulated end. |
| Raw P value | Nominal evidence before correction for thousands of tests. |
| Adjusted P value / FDR | Multiple-testing-adjusted evidence; primary threshold set at 0.05. |
| Venn overlap | Descriptive recurrence across thresholded study lists; not an effect-size meta-analysis. |
| Degree centrality | Number of network connections; not proof of causal importance. |